CLINICAL GUIDELINE

Early pregnancy bleeding

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Figure 1. Algorithm for initial management of early pregnancy bleeding

Figure 2. Algorithm for management of miscarriage

AUTHOR/S

This guideline has been developed by Monash Women's Gynaecology

TARGET AUDIENCE and SETTING

Monash Health medical staff and nurses.

Applicable to all women under the care of the Gynaecology services at Monash Health, with bleeding and/or pain in early pregnancy, and presenting for care in Emergency Departments (ED) or early pregnancy bleeding (EPB) clinics across Monash Health.

PRECAUTIONS

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DEFINITIONS

Early pregnancy: pregnancy in first trimester – up to 12 completed week's gestation.

Ectopic pregnancy: pregnancy outside the uterus, usually within fallopian tube, may also occur in abdomen, ovary, cervix or caesarean section scar.

Heterotopic pregnancy: intrauterine and extrauterine pregnancies occurring simultaneously.

Pregnancy of unknown location (PUL): positive pregnancy test but no pregnancy seen on ultrasound.

STANDARD REQUIREMENTS

When undertaking any clinical interaction with a patient, staff are expected to:

- Perform routine hand hygiene. Refer to the Hand Hygiene Procedure.
- Introduce yourself to the Patient and Carer/ Family if in attendance.
- Explain what you do and what you are there for
- Check patient identification. Refer to the Patient Identification Procedure.
- Obtain consent as per the Consent to Medical Treatment Procedure.
- Keep the patient/carer informed and involve them in decision making.
- Document interaction in the electronic medical record or health record using black pen; including date, time, signature and designation.

CLINICAL GUIDELINE

1. BACKGROUND

Early pregnancy bleeding refers to any bleeding in the first trimester, from confirmation of pregnancy via urine or serum human chorionic gonadotropin (hCG) up to and including 12 weeks completed gestation.

Approximately 15-20% of pregnant women will experience bleeding in the first trimester, and it represents the most common early pregnancy complication. 20% of recognised pregnancies will result in miscarriage.[1] This is not including biochemical pregnancies where a miscarriage occurs spontaneously prior to a woman recognising she is pregnant. In this instance, the signs of miscarriage can be mistaken for heavy or late menses.

Ectopic pregnancy affects 1-2% of pregnancies, but accounts for 75% early pregnancy mortality, and must be considered in any woman without a confirmed intrauterine pregnancy.[2]

Other causes of early pregnancy bleeding include implantation, lower genital tract lesions, infections, gestational trophoblastic disease.

Any woman of reproductive age with early pregnancy bleeding who presents to emergency with heavy bleeding or severe pain and/or haemodynamic instability requires urgent clinical assessment and medical review.

1. Evaluation

1.1. History

Take a detailed history of:

- Date of last period, regularity of periods, date of confirmation of pregnancy, course of the current pregnancy, gestational age, data from ultrasound, gravidity and parity.
- Past obstetrics and gynaecological history including cervical screening test status sexually transmitted infection and contraception use.
- Vaginal bleeding – quantify loss, e.g. how many pads changed over 1 hour.
- Pain; pelvic (unilateral or bilateral), cramping pains, shoulder tip pain, systemically unwell with subjective fevers/rigors/chills.

1.2. Examination

- Vital signs: hypotensive women with tachycardia can be experiencing vagal stimulation from cervical dilation associated with active miscarriage or from massive haemoperitoneum associated with ruptured ectopic pregnancy.
- Abdominal exam, assessing for tenderness, masses, signs of peritonism.
- Speculum exam:
  - Essential to assess amount of vaginal bleeding, and to view and remove any products of conception (POC) in cervix to prevent or treat cervical shock.
  - To visualise any lower genital tract lesions such as polyps or malignancy.
  - To collect swabs for suspected infection.

1.3. Investigations

Core investigations

- Full blood examination (FBE).
- Quantitative β-hCG.
- Blood group and antibody screen +/- Crossmatch as clinically indicated.
- Pelvic ultrasound
  - Discriminatory level of greater than 1500 international units/L required for visualisation of intrauterine gestation sac on transabdominal ultrasound.

Supplemental investigations

- Serum progesterone levels less than 35 nanomol/mL is more likely to indicate a potential non-viable or ectopic pregnancy.[3,4]

- Vaginal swab if suspected genital tract infection, or opportunistic screening for sexually transmitted infections for high-risk populations.
- Histopathology of products of conception passed.

1.4. General management principles

- Resuscitation and stabilisation of patient as clinically indicated.
- Treatment would depend on clinical situation, diagnosis, patient wishes.
- Send products of conception for histology in formalin, consider karyotyping if history of recurrent (≥2) miscarriages.
- Refer to Gynaecology Registrar on-call depending on clinical urgency.
- If major haemorrhage, refer to Massive Transfusion Protocol
- Early pregnancy bleeding (EPB) clinics are run daily on weekdays at Monash Medical Centre Clayton and Casey Hospital.
- Patients at Dandenong Hospital are to be referred to the Gynaecology Registrar on-call. Subsequent follow-up via outpatient β-hCG tracking as clinically indicated.

1.5. Rhesus (Rh) immunoglobulin (Anti-D) Immunoprophylaxis

All women who are Rhesus D negative, with no preformed Rhesus D antibodies are to be offered immunoprophylaxis with 250 international units of Rhesus D immunoglobin (Anti-D) following a sensitising event in the first trimester of a singleton pregnancy or 625 international units for a multiple pregnancy. This includes completed miscarriage (spontaneous, medical or surgical management), and ectopic pregnancy.

There is currently insufficient evidence to recommend Anti-D to women experiencing a threatened miscarriage.[5,6] However, where bleeding is heavy, repeated or associated with abdominal trauma, 250 international units of Anti-D is recommended in women with no preformed antibodies. See: Rhesus D negative women

2. Miscarriage

2.1. Definitions:

Missed miscarriage	Confirmed ultrasound diagnosis of non-viable pregnancy in asymptomatic patients without bleeding or pain, with no passage of POC, intact gestational sac in uterus.
Incomplete miscarriage	Some passage of POC, but some POC remaining within uterus
Complete miscarriage	Clinical assessment and/or previous ultrasound have confirmed intrauterine pregnancy, AND all the POC have been passed.

	Completeness is only definitely confirmed by ultrasound, but can be a presumptive diagnosis upon visualisation of apparently complete sac with placenta tissue expelled, and/or resolution of pain and bleeding.
Threatened miscarriage	Early pregnancy bleeding and/or abdominal pain where transvaginal ultrasound (TVUS) demonstrates viable intrauterine pregnancy, or intrauterine gestational sac that does not meet criteria for miscarriage.
Septic miscarriage	Any type of miscarriage, accompanied by evidence of intrauterine or systemic infection, urgent treatment is required with resuscitation, antibiotics and surgical evacuation of infected POC.

2.2. Diagnosis

Diagnosis is based on confirmed passage of products of conception or ultrasound findings consistent with the Australasian Society for Ultrasound in Medicine (ASUM) Criteria for miscarriage[7], as summarised below:

At initial or follow-up scan, an experienced practitioner using high-quality transvaginal equipment may diagnose pregnancy failure under either of the following circumstances:

- Mean sac diameter (MSD) greater or equal to 25 mm with no visible fetal pole; or
- Visible fetal pole with crown rump length (CRL) greater or equal to 7 mm but no fetal heart activity.

If no live embryo is demonstrated but the above criteria are not met, then the following criteria can be used to diagnose pregnancy failure by follow-up imaging:

- if the initial scan showed a fetal pole less than 7 mm with no fetal heart beat and a repeat scan in seven (7) or more days also shows no cardiac activity;
- if the initial scan showed an MSD greater or equal to 12 mm with no embryo and a repeat scan in seven (7) or more days does not show an embryo with cardiac activity;
- if the initial scan showed an MSD less than12 mm with no embryo and a repeat scan in fourteen (14) or more days shows no visible cardiac activity and the MSD has not doubled.

If there is any doubt as to the diagnosis of a miscarriage, offer a further scan.

2.3. Expectant management

Expectant management is suitable in cases of spontaneous and incomplete first trimester miscarriage in haemodynamically stable women. The greater the embryonic development at the time of demise, the less likely it is that expectant management will be successful.

- 70% of expectantly managed women will complete evacuation of POC within 2 weeks of diagnosis and 80% within 4 weeks.[8]
- There is low risk of infection (2-3%), similar to medical and surgical management.

- Higher risks of incomplete miscarriage, unplanned admissions (41%) and unplanned surgery (39%) when compared to medical (10% unplanned admissions, 30% unplanned surgery) and surgical management (8% unplanned admissions, 5% unplanned surgery).[9]
- If Rhesus negative, administer Anti-D 250 international units if below 12 weeks in a singleton pregnancy, 625 international units if multiple pregnancy or after 12 weeks.
- Women should have access to emergency facilities within 30 mins of residence
- Follow up is indicated if there is excessive blood loss, pain, fever or if POC have not been expelled after 4 weeks (or earlier if the woman changes her mind), at which time, consider surgical management.
- There are no contraindications to conceiving again after one menstrual cycle. However, it is important that the woman is emotionally ready before attempting another pregnancy.

2.4. Medical management

Regimen:

- Mifepristone 200 mg oral once (pre-treatment for missed miscarriages).
- 24 hours later followed by 400 micrograms misoprostol, buccal, sublingual or vaginally every 3 hours for 4 doses (Maximum 4 x 400 microg /day).
- Misoprostol is usually well tolerated with expected bleeding and pain, some adverse effects can include: fevers/chills, diarrhoea, vomiting. See: Misoprostol patient information
- Ideal for haemodynamically stable women wishing to avoid surgery, who will be compliant with medical regimen and follow-up, and is also close to medical facilities if she requires an unplanned surgical curettage:
  - Careful patient selection can reduce risks and optimise care for women electing medical management.
  - Patient selection must take into account social factors including access to 24 hour transport, ability to comply with treatment plan and follow-up, and availability of support person.
- For missed miscarriage, pre-treatment with mifepristone, followed by misoprostol resulted in higher rate of resolution at 3 days (84% vs. 67%), and a reduction in need for subsequent surgical management, with no difference in adverse events.[10,11]
  - Mifepristone is currently restricted to certified prescribers (ms2step.com.au)
- If Rhesus negative, administer Anti-D 250 international units if below 12 weeks in a singleton pregnancy, 625 international units if multiple pregnancy or after 12 weeks.
- Medical review is indicated at any time following medical management if there is excessive bleeding, pain or fever.
- Follow-up
  - If there is no bleeding within 24 hours of medical management, or if POC have not been expelled after 7 days of medical management, women ought to

represent to hospital for repeat dosage or counselling for other forms of management.

- Monitor serial weekly serum β-hCG until negative levels.

Exclusion criteria for medical management:

- haemoglobin less than 100 mg/dL
- known coagulopathy (or receiving anticoagulants)
- known or suspected ischaemic heart disease
- myocardial infarction
- severe asthma
- adrenal insufficiency
- living more than 1 hour travelling time from hospital.

2.5. Surgical management

Surgical management with suction dilatation and curettage provides the shortest time for complete evacuation of POC, and lowest rates of retreatment.[9]

- Counsel women regarding general anaesthetic risks, risks of infection, perforation (and potential need for laparoscopy and/or laparotomy if injury occurs), and Asherman's syndrome.
- If Rhesus negative, administer Anti-D 250 international units if below 12 weeks in a singleton pregnancy, 625 international units if multiple pregnancy or after 12 weeks.
- Pre-operative cervical ripening with 400 microg misoprostol 3 hours prior to surgery can assist in cervical dilation and avoiding cervical and uterine trauma.
- Send POC for histopathology. Consideration for karyotyping if recurrent pregnancy loss.

Indications for surgical management

- Heavy bleeding or pain and prolonged symptoms.
- Septic miscarriage.
- Suspected gestational trophoblastic disease.
- Failed expectant or medical management.
- Women with incomplete miscarriage within a reasonable period of time (2 weeks).
- Patient preference - may not be comfortable waiting for weeks to expel a non-viable pregnancy. When determining the timing of surgical intervention, the medical officer must consider the woman's psychological state, willingness to wait as well as the distance from hospital.

3. Ectopic pregnancy

Ectopic pregnancy (EP) is a pregnancy outside the uterus, usually in the fallopian tube (>95%), but can also occur in the abdomen, ovary or cervix as well as in a caesarean section scar. Ectopic pregnancies can also occur concurrently with an intrauterine pregnancy (heterotrophic pregnancy). Management of rarer non-tubal ectopic pregnancies (including interstitial EP) and heterotopic pregnancies must be referred for specialist care, and is outside the scope of this guideline.

3.1. Risk factors

- previous tubal surgery
- previous ectopic pregnancy
- previous STI/pelvic infection (PID)
- artificial reproductive technology including in-vitro fertilisation (IVF)
- documented tubal pathology
- smoker
- previous caesarean section.

3.2. Signs and symptoms

Ectopic pregnancy is initially asymptomatic and is easily missed as symptoms masquerade as a normal pregnancy. However, it remains the most common cause for early pregnancy mortality.

- Symptoms can include: abdominal pain (generally unilateral), vaginal bleeding (usually irregular, prolonged, rarely heavy bleeding), intra-abdominal bleeding can be gradual or catastrophic.
- Signs of ruptured ectopic pregnancy: distended silent abdomen with severe abdominal pain, shoulder tip pain, bulging cul-de-sac into posterior fornix of vagina, syncope, haemodynamic instability.

3.3. Diagnosis

- Positive serum β-hCG.
- Ultrasound showing an empty uterus, free fluid, adnexal mass (gestational sac with yolk sac, or live fetus or an inhomogeneous mass with associated free fluid ('blob' sign).

3.4. Expectant management

There can be a small carefully selected population of women with ectopic pregnancies where expectant management is appropriate, as suggested by NICE Guidelines.[12]

Expectant management could be considered for women who:

- Are clinically stable, pain-free; **AND**
- have a tubal ectopic pregnancy measuring less than 35 mm with no visible heartbeat on transvaginal ultrasound scan **AND**

- have serum hCG levels below 1,500 international units /L **AND**
- are able to return for follow-up via EPB clinic.
  Note: If planning follow-up with GP, they must be contacted and agree to plan, and be provided with a clear escalation plan.

Repeat serum β-hCG levels must be performed on days 2, 4 and 7 after the original test and:

- If β-hCG levels drop by 15% or more, from the previous value on days 2, 4 and 7, then repeat weekly until a negative result is obtained.
- If β-hCG levels do not fall by 15%, stay the same or rise from the previous value, review the woman's clinical condition and consider alternative forms of management.

3.5. Medical management

Medical management of EP with methotrexate can successfully treat haemodynamically and clinically stable women with EP.

Prior to administration, all eligible women shall be counselled and formally consented regarding:

- risks/adverse effects of methotrexate
- possible failure and requirement for surgical management
- need to avoid pregnancy for 3 months following treatment
- If Rhesus negative, administer Anti-D 250 international units if below 12 weeks in a singleton pregnancy, 625 international units if multiple pregnancy or after 12 weeks.
- Methotrexate for ectopic pregnancy patient information

See: Methotrexate for ectopic pregnancy Procedure

Eligibility criteria:

- Clinically stable with no signs of peritonism.
- Compliant, able and willing to attend for follow-up scans and blood tests.
- Early unruptured tubal ectopic pregnancy as indicated by:
  - under 8 weeks gestation
  - serum β-hCG levels less than 5000 international units
  - ectopic sac up to 5 cm
  - no fetal heart pulsation.
- Surgery contraindicated or likely to be complicated.
- Non-tubal ectopic pregnancy: cornual/cervical/caesarean scar.

Note that patients outside of eligibility criteria can still be suitable for treatment, but this must be discussed with the Reproductive Medicine Unit Fellow or the Gynaecology Consultant.

Contraindications:

- Haemodynamic instability.

- Haemoperitoneum (suspected on physical examination, ultrasound).
- Severe immunodeficiency.
- Pre-existing liver disease, haematological abnormalities (leukopaenia, thrombocytopaenia), pulmonary disease, or renal disease.
- Sensitivity to methotrexate.
- Poor access to hospital services/unable to comply with treatment plan and follow-up.

Potential adverse effects:

- Mild stomatitis, gastritis, enteritis, photosensitivity, diarrhoea, transient elevation in liver enzymes (in 3%).
- Rare side effects: alopecia, bone marrow suppression, hepatotoxic effects (2 case reports only).

3.6. Surgical management

Surgical management with laparoscopic salpingostomy or salpingectomy is the mainstay of treatment for ectopic pregnancy.

- Consider laparoscopic salpingostomy (tubal conservation) where there is known contralateral tube damage, as there is a known higher cumulative pregnancy rate in this group if undergoing salpingostomy.[13]
- Rates of future fertility are similar whether salpingostomy or salpingectomy are performed.
- Send POC for histopathology and if no confirmed tissue diagnosis (or if a salpingostomy was performed), women must have weekly bHCG tracking until negative.

4. Management of pregnancy of unknown location (PUL)

Pregnancy of unknown location occurs when a woman has a positive urinary or serum hCG without a gestational sac on ultrasound.

PUL is not a diagnosis, and requires further investigations and strict follow-up until diagnosis or pregnancy is resolved.

PUL may occur due to early stages of pregnancy below the discriminatory serum β-hCG level of 1500 international units /L, above which a gestational sac should be visualised on transvaginal ultrasound.

Correlation of serum β-hCG levels with ultrasound findings provides the most diagnostic significance:

- Failure to double in 48 hours predicts pregnancy failure but does not distinguish between an intrauterine spontaneous miscarriage and an ectopic pregnancy.
- If the β-hCG is around 1,500 international units /L a gestational sac should be visible on transvaginal ultrasound.

- If the intrauterine gestational sac is not visible by the time the β-hCG is at or above this threshold, the pregnancy has a high likelihood of being ectopic, indicating the possible need for a laparoscopy

Figure 1. Algorithm for initial management of early pregnancy bleeding

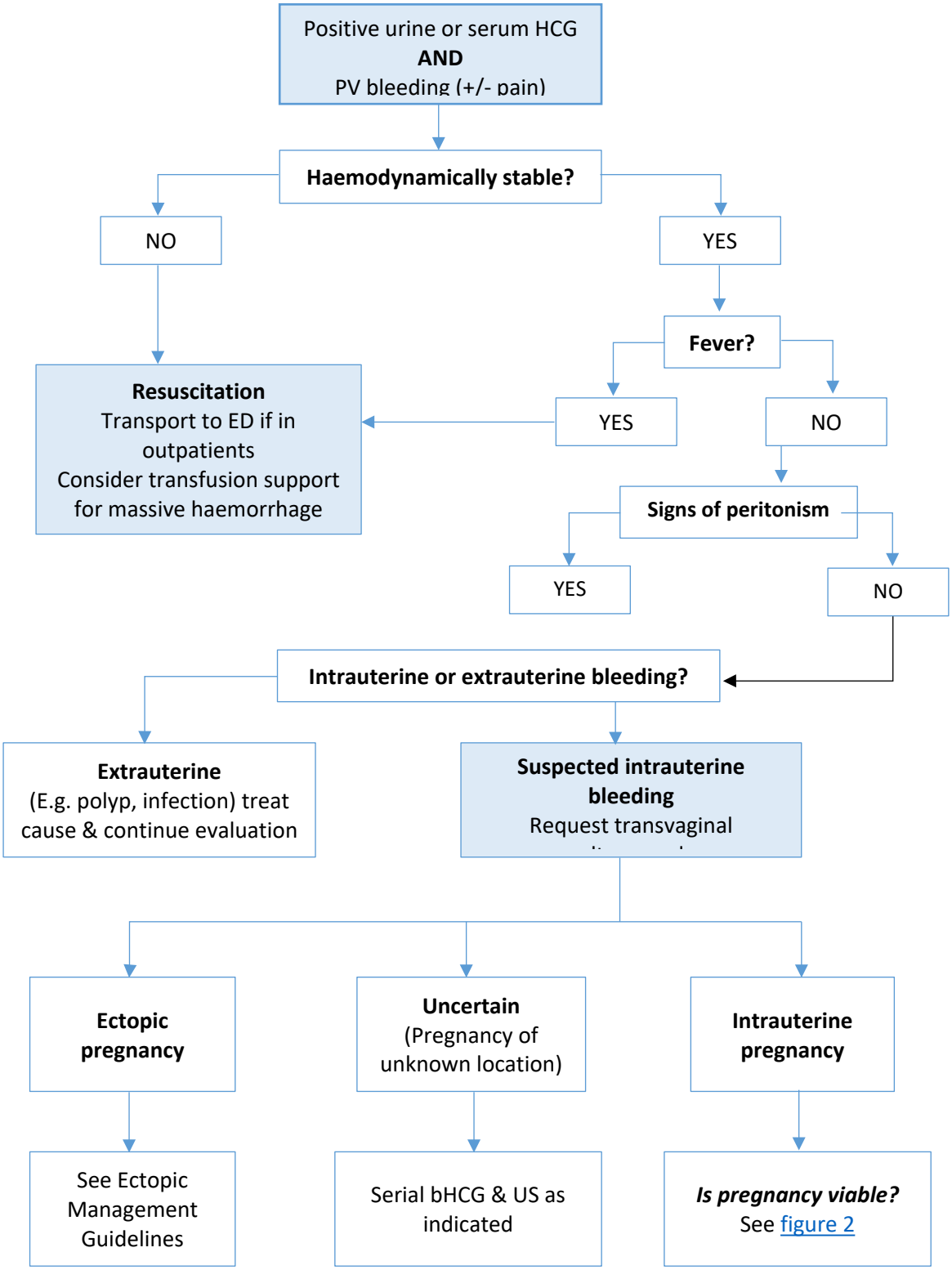
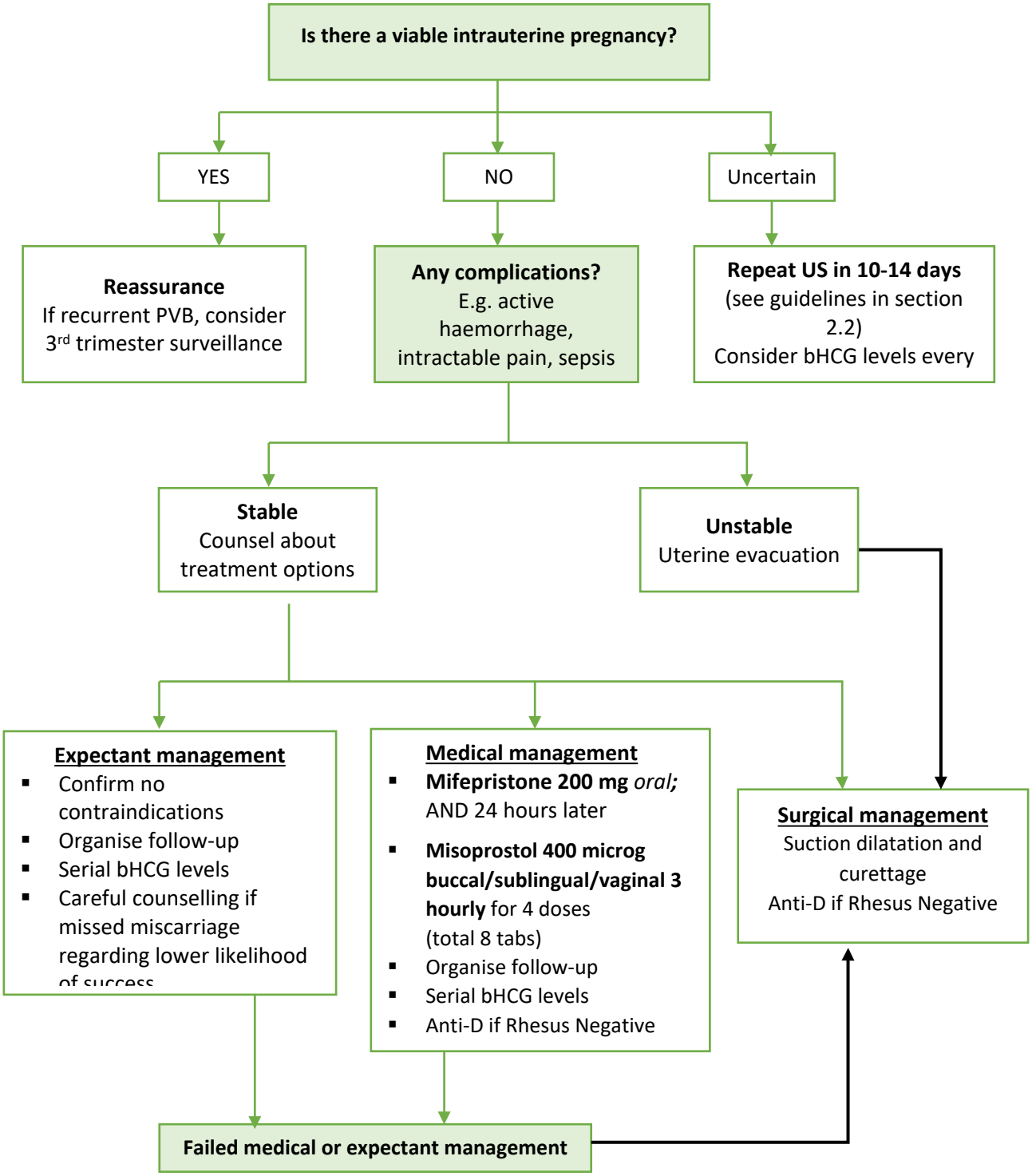


Figure 2. Algorithm for management of miscarriage



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